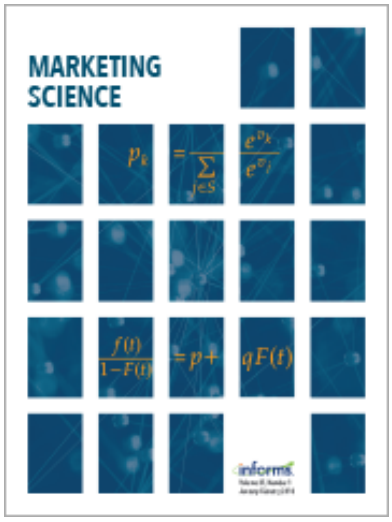
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Mass Shootings and Their Impact on Retail

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Abstract. Mass shootings in the United States have become more frequent, often targeting retail locations such as shopping malls that attract high foot traffic. We combine detailed data on mass shootings with debit and credit card transaction data at individual stores to assess the economic impact of mass shootings. Using a difference-in-differences framework, we find that affected stores experience lower demand, smaller order sizes, decreased foot traffic, and reduced customer dwell times, resulting in a significant decline in retail revenues. We also find the impact of mass shootings diminishes with distance, adversely affecting stores within a radius of up to 1.25 miles. Additionally, we examine the likelihood of store closures after incidents and factors contributing to store survival. Overall, we estimate the annual economic cost of mass shootings at $27 billion in lost revenues to retail businesses. Consistent with consumers' fear and safety concerns, we observe heterogeneous effects by business type (e.g., nonessential stores are more impacted than essential stores), channel substitution from in-person to online shopping, and moderation by local exposure to gun-related violence.

History: Catherine Tucker served as the senior editor.

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Supplemental Material: The online appendix and data files are available at https://doi.org/10.1287/mksc.2024.0752.

Keywords: retailing • gun violence • natural experiment • staggered difference-in-differences • causal inference

1. Introduction

Mass shootings in the United States have become more frequent in recent years. Between 2016 and 2019, the country averaged 371 mass shootings per year. This number rose to 650 annually between 2020 and 2023.[1] The growing prevalence of mass shootings presents challenges for retail businesses. First, retail locations such as shopping malls are common targets due to their high foot traffic,[2] making them the most common setting, followed by restaurants and bars.[3] Second, nearly two-thirds of retailers have identified mass violence or active assailant events as a growing concern (National Retail Federation 2023). In response to the recent uptick in mass shootings, some public companies have started assessing and disclosing the potential financial impact of these events in their annual reports.[4] Although the human toll of these events is evident, their economic impact on retail businesses is not known. We present the first systematic investigations into the economic cost of mass shootings to retail businesses and how they impact consumers.

We combine U.S. mass shooting incident data with store-level credit and debit card transaction data that are matched to the location of the incident. Because mass shootings are salient and plausibly random events, we implement a difference-in-differences (DID) strategy around the timing and location of mass shootings to estimate their causal impact on consumer shopping behavior. Specifically, we compare the performance of stores that encountered a mass shooting (i.e., treatment group) to that of stores that did not (i.e., control group).

We examine the record of all mass shootings in the United States that occurred at or near a retail location between January 2022 and June 2023, totaling 486 incidents. By tracking the performance of 16,431 individual retail businesses over a 30-month period, our DID estimates show that, on average, affected stores experienced a 12.9% reduction in transactions and a 4.8% decrease in order size, leading to an overall 18.7% decrease in revenue. These findings are corroborated by foot traffic data, which show significant reductions in both foot traffic and dwell time at the affected stores

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after incidents. When examining store closures due to mass shootings, we find that 16.7% of affected stores shut down. However, certain types of stores, such as branded and franchised stores, are less likely to close.

We also estimate how the impact of mass shootings varies with distance from the incident. We show that mass shootings impact not only directly affected stores but also businesses within a 1.25-mile radius. The impact, however, diminishes with distance: stores closest to the incident (within 0.1 miles) face an average revenue decline of 24.6%, whereas those further away (between 1 and 1.25 miles) see a smaller decline of 8.7%. Beyond 1.25 miles, the effect is not statistically significant.

Accounting for store closures and using DID estimates based on the proximity of stores to incidents, we estimate the nationwide economic cost of mass shootings at $27 billion annually in forgone revenue for affected retailers, equivalently erasing 0.5% of the U.S. retail gross domestic product (GDP) each year.

The decrease in foot traffic and dwell times at impacted stores following an incident suggests that heightened consumer fear and safety concerns may lead consumers to avoid or curtail time spent there. Further supporting this, we observe heterogeneous effects by business type, with nonessential stores more affected than essential ones, evidence of channel substitution from in-person to online shopping, and moderation by local exposure to gun-related violence, where stores in areas with higher firearm mortality experience smaller impacts.

We conduct a battery of additional analyses to check the validity and robustness of our results. First, we use recent advances in staggered DID (Callaway and Sant'Anna 2021) to account for the variation in the treatment timing of mass shootings. We also find consistent results across various functional form assumptions and sample selection criteria.

We offer several managerial and policy implications. First, this study is the first to document and quantify the economic cost of mass shootings on retail businesses, providing benchmarks for understanding these risks. Second, our findings show that the impact of mass shootings extends beyond directly targeted stores, affecting businesses within a 1.25-mile radius. With the increasing number of mass shootings, our findings offer a novel perspective on the national discourse by highlighting the geographically widespread economic impacts on businesses and underscoring the need for attention from policymakers to address it. Finally, we demonstrate the use of online channels in bolstering store resilience against adverse shocks such as mass shootings.

2. Related Literature

Our study adds to three streams of literature. First, our study contributes to the literature examining the effect of mass shootings on consumer psychology. Hughes et al. (2011), Soni and Tekin (2023), and Lagerborg et al. (2023) highlight mass shootings' prolonged negative impact on the community and individual emotional well-being, as well as consumer confidence levels. Studies have also shown that fear of mass shootings reduces leisure activities (Clark and Stancanelli 2017) and discourages consumers from attending public events or going to the mall (American Psychological Association 2019). However, these studies primarily rely on surveys. In contrast, we employ detailed data on customer visits and spending at retail stores to investigate the impact of mass shootings on businesses.

Second, our study builds on literature examining the financial and economic consequences of mass shootings. Brodeur and Yousaf (2025) and Lagerborg et al. (2023) report the adverse effect of mass shootings on local economies, including increased unemployment rates and decreased industrial production. Other studies, like Gopal and Greenwood (2017), Karan (2019), and Sakariyahu et al. (2023), focus on repercussions for financial markets. Shaik et al. (2023) study the economic impact of fatal school shootings on grocery purchases and home values. Although these studies primarily consider broader macroeconomic indicators at the county or national levels, our study takes a different approach by examining the performance metrics of individual retail stores using location-based credit and debit card transaction data.

Finally, our study contributes to policy discussions on gun violence and the availability of firearms. Luca et al. (2020) find that mass shootings have an outsized influence on policy responses, leading to a 15% increase in firearm bills introduced in a state in the following year. Recent research, such as Balakrishna and Wilbur (2022), Kim and Wilbur (2022), and Moshary et al. (2023), investigates the firearm market by analyzing sales and consumer demand for firearms. Our study adds to this dialogue by quantifying the economic cost of mass shootings.

3. Data

We utilize three main data sets: (1) mass shooting data from the Gun Violence Archive (GVA), (2) debit and credit card transactions data from SafeGraph Spend, and (3) foot traffic data from Advan Patterns.

3.1. Mass Shooting Data: GVA

We obtain data on mass shooting incidents from the GVA,[5] a not-for-profit, independent research group. The GVA collects daily data from more than 7,500 sources, including law enforcement, media, government, and commercial sources, providing near-real-time information on gun violence incidents in the United States. Following the GVA's definition, a mass

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shooting is defined as an incident in which four or more people are shot, excluding the shooter. For each incident, the archive catalogs the date, location, and number of fatalities and injuries.

3.2. Location-Based Retail Spending Data: SafeGraph Spend

SafeGraph Spend collects anonymized debit and credit card transactions from a panel of around 10.6 million consumers, spanning more than 1.1 million business locations in the United States.[6] Each transaction is geocoded and linked to a brick-and-mortar business location. The data provide a spatially granular view into American commerce, allowing us to see how much customers spend at each place and month.[7] Online Appendix C.1 offers additional background information on SafeGraph Spend and highlights its representativeness by comparing it to U.S. Census statistics, including the number of customers and income levels.

SafeGraph Spend provides business names, precise locations, and monthly performance metrics for each establishment. We focus on three performance measures at the store-month level: (1) *Total Revenue*, representing the total dollar amount spent by customers; (2) *Number of Transactions*, indicating the total number of credit and debit card transactions; and (3) *Order Size*, measured as the median dollar amount spent per transaction.[8]

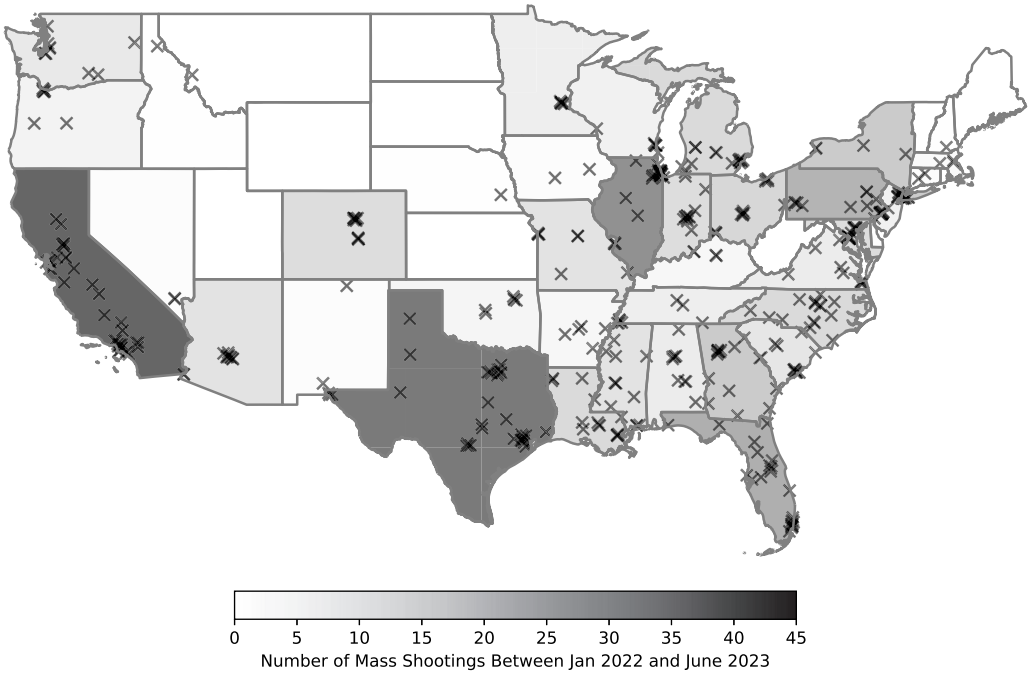
We categorize stores using the North American Industry Classification System (NAICS) codes, covering businesses such as restaurants, bars, grocery stores, gas stations, clothing stores, and pharmacies. Tables C.1 and C.2 in the Online Appendix provide a list of these categories using four-digit and six-digit NAICS codes, respectively, along with the count of stores in our data.

3.3. Estimation Sample Construction

This section outlines how the GVA and SafeGraph Spend are merged to identify stores affected by mass shootings (i.e., treated stores), followed by details on finding comparable stores (i.e., control stores) via a matching process.

3.3.1. Sample of Mass Shootings. From the GVA, we obtained all mass shooting incidents in the United States between January 2022 and June 2023 that occurred within 0.25 miles of a retail store in SafeGraph Spend.[9] This resulted in a total of 486 incidents, which we term retail mass shootings: incidents occurring within walking distance of a retail location. Figure 1 depicts the geographical distribution of these incidents, indicating that mass shootings are frequent in population centers. In terms of the site of mass shootings in our sample, retail/commercial was the most frequent incident location (41.7%), followed by residential (25.1%) and mixed-use commercial/residential (23.4%). Our sample also includes other incidents, such as school shootings, drive-by shootings, shootouts, robberies, and domestic violence. Table C.3 in the Online Appendix provides a more detailed characterization of these incidents.

Figure 1. Map of Retail Mass Shootings Between January 2022 and June 2023



Notes. This figure shows mass shooting incidents in our sample that occurred between January 2022 and June 2023. Each × on the map denotes the location of a mass shooting incident.

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Our sample covers several high-profile retail mass shootings, such as the shooting at a Tops Friendly Supermarket in Buffalo, NY, that killed 10 and injured three; the shooting at the Allen Premium Outlets that killed nine and injured seven; and the shooting at Greenwood Park Mall in suburban Indianapolis that killed four and injured two.

On average, a mass shooting results in 0.97 deaths (excluding the shooter) and 4.29 injuries, with the median values of one death and four injuries in our sample.

3.3.2. Treated Stores and Descriptive Evidence. Based on our sample of mass shootings, 486 incidents occurring within walking distance of 0.25 miles from a retail store, we then identified 3,323 affected (treated) stores within 0.25 miles of these incidents. We identify these stores by calculating the distances between the stores and the locations of mass shooting incidents, using the coordinates provided in the two data sets.

Our SafeGraph Spend data cover the period from July 2021 to December 2023. By examining mass shooting incidents between January 2022 and June 2023, we ensure that each treated store has at least six months of pre- and posttreatment observations.

Table 1 provides the summary statistics of the treated stores' retail performances. By comparing the average values before and after mass shooting incidents, we observe that the total revenue decreases by 13% after the incident. Similarly, the average number of transactions and the order size decreases by 9.6% and 6.8%, respectively. Figure A.1 in the Online Appendix, which plots the trend of average weekly revenues before and after a mass shooting incident, also exhibits a noticeable decrease after an incident.

3.3.3. Comparable Control Stores via Matching. To facilitate a DID estimation of the causal impact of mass shootings, we identify unaffected comparable control stores using the following steps.

First, we define the pool of eligible control units to be stores located between 10 and 94.7 miles from the incident location. The minimum distance of 10 miles is to ensure that the control stores are not too close to risk spillover effects from the treated stores,[10] whereas the maximum distance of 94.7 miles is to ensure similarity in the underlying macroeconomic conditions for each control and treated pair. Second, for each treated store, we employ exact matching to find the set of control stores of the same business type (based on NAICS). For instance, a treated pharmacy is matched with other pharmacies as its control group. Third, for each treated store, we apply Mahalanobis distance matching to find five control stores that are most similar in terms of performance measures preshooting.

The final sample is a balanced panel data of 16,431 retail stores (3,323 treated and 13,108 control stores) between July 2021 and December 2023, totaling 492,930 store-month observations. Online Appendix B provides further details and justification of the matching procedure and demonstrates a covariate balance between treated and control stores. Online Appendix D.2 discusses the robustness of our results with respect to alternative matching methods.

3.4. Foot Traffic Data

We augment our stores' transaction data with foot traffic data from Advan Patterns.[11] These data provide monthly visit counts to places and points of interest based on a panel of mobile devices, including visit frequency and duration. In particular, it shares the same store identifier with SafeGraph Spend, allowing us to examine the same set of treated stores over the same time period (balanced panel). We focus on the following store-month outcomes: (i) *Visits*, representing the total number of visits to the location, and (ii) *Dwell Time*, indicating the median duration of visits (in minutes).

4. Empirical Analyses

We separately analyze the effect of mass shootings on two types of stores: (1) operating stores and (2) closed stores. We consider a store as *closed* if we observe three or more consecutive months of closure after the incident and *operating* otherwise. Of the 3,323 treated stores, 554 (16.7%) are closed, with the remainder being operating stores.

First, we analyze the effect of mass shootings on operating stores' performance metrics and foot traffic (Section 4.1) and examine store closure (Section 4.2). We then estimate how the effect decays over geographic

Table 1. Descriptive Statistics of Treated Stores

Outcome	Before mass shooting Mean (standard error)	After mass shooting Mean (standard error)	Difference (%)
Total revenue ($)	6,131.5 (95.2)	5,336.7 (84.8)	−12.96
No. of transactions	179.79 (2.1)	162.50 (2.0)	−9.62
Order size ($)	33.35 (0.3)	31.07 (0.3)	−6.83

Notes. We winsorized the raw data at the 99.9th percentile for this table. The total number of treated observations is 99,690, that is, 3,323 treated units over 30 months. Of these observations, 43,944 are preperiod, and 55,746 are postperiod. SafeGraph's panel samples a fraction of the store's total revenue and thus does not reflect the total revenue.

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distance (Section 4.3). Finally, we arrive at an estimate of the economic cost of mass shootings to the retail industry in Section 4.4.

4.1. Main Effect

Our main empirical specification is a two-way fixed-effect (TWFE) DID regression:

$$Y_{it} = \beta \cdot PostTreat_{it} + \delta_i + \tau_t + \epsilon_{it}. \quad (4.1)$$

The dependent variable Y_{it} represents the outcome measure of store i in month t, specifically the logarithmic functions of total revenue, the number of transactions, order size, foot traffic, and median dwell time. The variable $PostTreat_{it}$ equals one if store i is affected by a mass shooting incident and if t is after the incident; otherwise, it is zero. The coefficient β is our parameter of interest. We include fixed effects for stores, δ_i, and time periods, τ_t, to account for store-specific and macroeconomic factors that may affect the outcome. Standard errors are clustered at the incident level to account for correlations among the error terms of different stores affected by the same mass shooting incident.

The standard TWFE DID framework assumes that treatment occurs at the same time for all treated units. However, mass shootings occur at different times, resulting in staggered treatment timing. As an alternative model, we implement the staggered DID estimator of Callaway and Sant'Anna (2021) (CS).

Both the TWFE and the CS estimates show that mass shootings have a significant negative effect on revenue, number of transactions, and order size (Panel A in Table 2). In particular, affected stores on average experience a 19% reduction in revenue, 13% lower transactions, and 5% smaller order size, compared with the control stores. The magnitudes of coefficients from both TWFE and CS are comparable.

In Panel B of Table 2, we examine the effect of mass shootings on consumer visits and their median dwell time. We find that mass shootings significantly decrease foot traffic to affected stores by 23% (column 7 of Table 2). Additionally, the median dwell time of visitors at these stores decreases by 18%. The reduced dwell time explains the negative impact of mass shootings on both intensive and extensive margins: with less time spent in stores, individuals purchase fewer items per visit, leading to the smaller order sizes observed in Table 2.[12]

We now examine how the impact of mass shootings varies over time, relative to the month of the incident. We conduct an event study using a dynamic TWFE regression and plot the parameter estimates in Figure 2. The preincident coefficients indicate no violation of the parallel trends assumption. Meanwhile, the negative postincident coefficients demonstrate the long-lasting adverse effect of mass shootings on store revenues. In Figure A.2 in the Online Appendix, we further extend the event study to different store types, considering factors such as store closures and proximity to the incident location and assessing the recovery dynamics of different types of stores.

Lastly, we conduct extensive robustness checks to validate our estimates. The results remain robust to (1) using higher frequency data: daily and weekly revenue (Table D.2 in the Online Appendix), (2) employing control stores located at least 20 miles from the incident and adjusting the number of matched controls for each treated unit (Tables D.3 and D.4 in the Online Appendix), (3) conducting placebo tests (Table D.5 in the Online Appendix), (4) accounting for industry-sector-specific

Table 2. Overall Effect of Mass Shootings on Retail Stores

Panel A: Effect on stores' performance

	(I) TWFE			(II) Callaway and Sant'Anna		
Variables (log)	Revenue (($) US Dollar) (1)	Transactions (2)	Order size ($) (3)	Revenue ($) (4)	Transactions (5)	Order size ($) (6)
PostTreat	−0.187***	−0.129***	−0.048***	−0.183**	−0.123**	−0.049**
	(0.026)	(0.023)	(0.005)	(0.015)	(0.013)	(0.007)
Observations	425,730	425,730	425,730	425,730	425,730	425,730

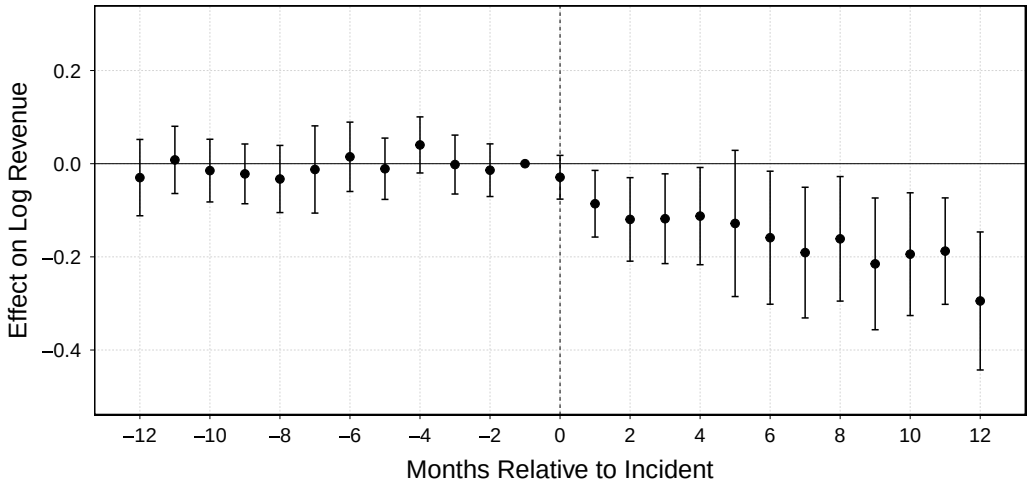
Panel B: Effect on foot traffic and dwell time

	(III) TWFE		(IV) Callaway and Sant'Anna	
Variables (log)	Visits (7)	Dwell time (8)	Visits (9)	Dwell time (10)
PostTreat	−0.231***	−0.182***	−0.232***	−0.169***
	(0.036)	(0.038)	(0.022)	(0.023)
Observations	376,170	371,982	376,170	371,982

Notes. All models include store and time fixed effects. Standard errors in parentheses are clustered at the incident level for TWFE and bootstrapped for Callaway and Sant'Anna.
***$p < 0.01$; **$p < 0.05$.

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Figure 2. Dynamic Event Study



Notes. This figure shows the event study corresponding to the main TWFE specification (Equation (4.1)). The event study using Callaway and Sant'Anna is provided in Figure D.1 in the Online Appendix, with Table D.1 showing that both event studies provide comparable estimates. Standard errors are clustered at the incident level. The reference relative period is −1. We restrict to stores with a minimum of 12 months of pre- and postperiods and group the endpoints at −12 and 12.

and state-specific time trends (Table D.6 in the Online Appendix), (5) considering the effect of COVID-19 (Table D.7 in the Online Appendix), (6) excluding the small number of stores affected by multiple incidents, and (7) distinguishing stores located within complexes such as malls or plazas (Table D.9 in the Online Appendix).

4.2. Store Closure

Following our analysis of the impact on operating stores, we now turn our attention to closed stores. We examine the factors influencing the likelihood and duration of store closure among the treated stores using two dependent variables: (1) $Closed_i$, indicating whether store i is observed to be closed for three or more consecutive months following a mass shooting, and (2) $ClosedMonths_i$, representing the total months of closure (which may be nonconsecutive).

We find that branded stores and stores with a larger revenue are less likely to experience closure (Table A.1 in the Online Appendix). For example, being part of a national franchise or major brand[13] reduces the probability of closure by 0.102 and shortens closure duration by 1.3 months. This resilience may be due to higher brand equity and better resources to handle shocks like mass shootings. Branded stores also perform better in terms of revenue, number of transactions, and order size postincident (Table A.5 in the Online Appendix). These findings are consistent with franchising literature, which reports higher survival rates for franchised stores compared with independent ones (Lafontaine et al. 2019).

On the other hand, *Nested* stores and those with higher order sizes are associated with a greater likelihood of store closure. *Nested* stores, defined as those located within or forming part of larger structures such as shopping complexes, are often enclosed spaces with limited escape routes, which may be perceived as less safe during mass shootings. Supporting this, foot traffic to *Nested* stores declined more sharply after mass shootings (Table A.6 in the Online Appendix). Similarly, stores with higher order sizes, often nonessential retailers such as jewelry, luggage, shoe, and electronics stores in our sample, are also more prone to closure.[14]

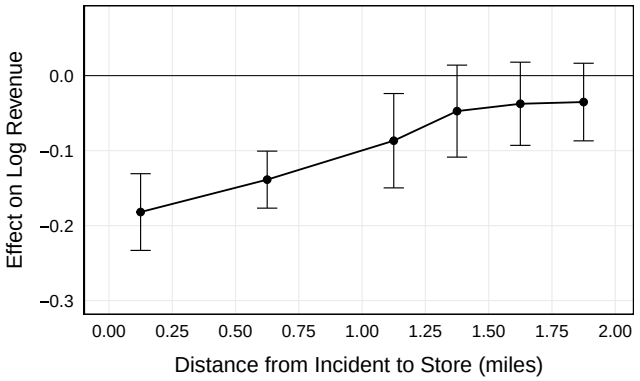
4.3. Effect over Distances

Our main sample consists of stores located within 0.25 miles of an incident, corresponding to the immediate vicinity of mass shootings, where we observe an overall negative impact. Splitting the treated stores into three equal-sized distance groups further reveals diminishing effects with distance: Stores within 0.11 miles experience the largest impact at –25%, followed by a –16.4% impact for stores 0.11–0.19 miles away, and a –15% impact for those 0.19–0.25 miles away (see column 1 of Table A.2 in the Online Appendix).

To further assess the spatial extent of the impact, we expand our sample to include stores within 2 miles of an incident, matching control stores in the same way as in our main sample. Figure 3 plots the estimates of the group-specific treatment effect. We find that the spatial impact of mass shootings diminishes over distance, becoming statistically insignificant beyond a 1.25-mile radius. This pattern also justifies our choice of the control group, consisting of stores located at least 10 miles away from any incident, which avoids spillovers from treated to control units. We report the full results in Table A.2 in the Online Appendix.

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Figure 3. Spatial Impact of Mass Shootings



Notes. We construct dummy variables for whether a treated store's distance from its incident location is within the following distance groups: $(0, 0.25]$, $(0.25, 1]$, $(1, 1.25]$, $(1.25, 1.5]$, $(1.5, 1.75]$, $(1.75, 2]$ miles. The plot shows estimates of the group-specific treatment effect (Table A.2 in the Online Appendix) and their 95% confidence intervals. Standard errors are clustered at the incident level. We position the estimates of each distance group at their respective midpoints on the x axis, for example, the coefficient for the group with distance between 1 and 1.25 miles is positioned at 1.125.

4.4. Economic Cost

This section quantifies the total economic losses to the retail industry due to mass shootings. We begin by calculating the aggregate economic loss for stores within our sample (in-sample loss) as follows:

$$\sum_{i \in \text{Treat}} \left\{ \left[(1 - \text{closure}_i) \times \underbrace{\text{mean revenue}_i}_{\text{in pre-period}} \times \underbrace{\text{DID estimates}}_{\text{from Figure 3}} \right] \right. \tag{4.2}$$

$$\left. + \left[\text{closure}_i \times \left(- \underbrace{\text{mean revenue}_i}_{\text{in pre-period}} \right) \right] \right\}. \tag{4.3}$$

For operating stores (Equation (4.2)), we calculate their economic losses using the distance-specific DID estimates from Figure 3, aggregating across all stores within a 1.25-mile radius while accounting for diminishing impacts with increasing distance from the incident. For closed stores (Equation (4.3)), losses are calculated based on their preperiod revenues.[15] We then scale this in-sample loss with SafeGraph's sampling rate and store coverage ratio (see Online Appendix A.3 for details).

The estimated economic loss is $27 billion annually in foregone revenue to the affected retailers. To put this number in context, the retail industry's total GDP contribution was $5.3 trillion, accounting for 20.4% of the U.S. GDP in 2022.[16] Hence, the loss from mass shootings amounts to approximately 0.5% of the U.S. retail GDP annually.

This estimate represents the loss from mass shootings occurring within a single year, based on the incident rate observed during our sample period from January 2022 to June 2023. If incident rates were to increase (or decrease), the corresponding annual loss would rise (or fall) accordingly. Furthermore, this figure reflects only counterfactual revenue losses and does not take into account direct costs such as facility repairs and medical expenses associated with shooting incidents.

5. Consumer Fear and Safety Concerns: Evidence

Because our effect is long-lasting, temporary disruptions such as police investigations, media presence, or scene clean-up and repair are unlikely to explain our findings. A more plausible mechanism is heightened fear and safety concerns, which prompt consumers to avoid or curtail time spent in affected stores (American Psychological Association 2019). This section presents evidence showing that fear and safety concerns drive consumers to avoid visiting or spending time at physical retail locations impacted by mass shootings.

5.1. Heterogeneous Effects by Business Types

In Section 4, we saw that mass shootings negatively impact foot traffic and dwell time at affected locations. Additionally, if fear and safety concerns were to deter prolonged stays at affected locations after mass shootings, businesses that rely on longer customer dwell time may experience relatively greater impacts. To investigate this, we examine two retail categories with long and short dwell times: sit-down versus fast-food establishments. Sit-down establishments, including bars, full-service restaurants, and coffee shops, encourage consumers to stay longer. Conversely, fast-food establishments, such as fast-food chains and fast-casual restaurants, are designed for shorter visit durations.[17] In column 1 of Table 3, we show that sit-down venues are more adversely affected than fast-food establishments.

Consumer responses to mass shootings may also vary based on the nature of the products sold in a store—specifically, whether the store offers essential products meeting basic needs or nonessential products with discretionary demand that can be postponed or avoided. We examine this by looking at retail stores and categorizing them into either essential or nonessential. For instance, essential stores are groceries, pharmacies, gasoline stations, and convenience stores, whereas nonessential stores sell clothing, shoes, electronics, jewelry, and sporting goods. This analysis excludes food-service establishments, such as restaurants and bars, which are analyzed separately in the previous section.[18]

We find that essential stores experience less impact from mass shootings compared with nonessential stores (column 2 of Table 3). Therefore, although there

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Table 3. Evidence of Consumer Fear as a Driving Factor

	Heterogeneity		Channel substitution		
	Revenue ($)		Online ($)	Offline ($)	Total ($)
Variables (log)	(1)	(2)	(3)	(4)	(5)
PostTreat	−0.079**	−0.337***	0.093**	−0.207***	−0.146**
	(0.036)	(0.064)	(0.045)	(0.039)	(0.027)
PostTreat × *Sitdown*	−0.127***				
	(0.040)				
PostTreat × *Essential*		0.194***			
		(0.067)			
Observations	242,370	183,360	193,170	193,170	193,170

Notes. The first column considers only food-related establishments. The second column considers all non−food-related establishments. The third to fifth columns consider restaurants only. All models include store and time-fixed effects. Standard errors in parentheses are clustered at the incident level.

***$p < 0.01$; **$p < 0.05$.

is a general tendency to avoid affected areas, this effect is less pronounced for essential shopping.

Overall, our results corroborate the mechanism that consumer response is driven by safety concerns and the fear of spending time at locations that have experienced mass shootings.

5.2. Online Channel Substitution

Following mass shooting incidents, consumers may perceive affected locations as unsafe. As a result, consumers may shift from in-person shopping to online ordering to minimize the perceived risks associated with physical visits.

To examine this shift, we focus on restaurants, defined as Full-Service Restaurants (722,511) and Limited-Service Restaurants (722,513) in the NAICS, for two reasons. First, restaurants are frequent targets in mass shootings, making them the most common retail location type in our sample, comprising 43% of all treated units.[19] Second, unlike other businesses, our data provide comprehensive insights into online channel orders for restaurants.[20]

We find that restaurants affected by mass shootings experience a significant increase in online revenue (+9.3%), as we show in column 3 of Table 3. However, because mass shootings negatively impact in-person revenue to a larger extent (column 4), the overall total revenue still decreases (column 5). Table A.7 in the Online Appendix further shows positive effects on online order size and the number of transactions, indicating substitution from offline to online channels along both the intensive and extensive margins.

Overall, our findings that consumers are shifting from offline to online channels support the notion that consumers seek to curtail the time spent at the affected locations. From a managerial perspective, the presence of an online channel can help businesses mitigate losses, thus making them more resilient to negative shocks.

5.3. Desensitization and Normalization to Gun Violence

Consumer fear and safety concerns from these incidents may be influenced by local conditions and experiences. Previous studies (Li et al. 2017, Goldstein and Prater 2023) reported evidence of desensitization and normalization to recurrent acts of violence, suggesting that consumers in areas with higher baseline firearm death rates may exhibit smaller changes in shopping behavior following mass shootings. We find the impact of mass shootings is significantly moderated in counties with higher firearm death rates, suggesting some desensitization to gun-related violence in these areas (see Table A3 in Online Appendix A.4).

5.4. Supply-Side Factors

In addition to consumer fear, supply-side factors may also contribute to the decline in store performance following mass shootings. Staffing issues after such events may lead to store closures or reduced operating hours. Moreover, heightened fear and lowered morale among employees may result in declining service quality toward customers, further impacting consumer demand and revenue. Although these supply-side factors could alter the interpretation of our findings, they should not change our estimate of the economic cost.

Finally, stores might increase promotions to attract consumers or implement enhanced security measures to alleviate consumer fears after mass shootings. As these efforts are likely designed to boost revenue, failing to account for these factors could lead to an underestimation of the true impact. In other words, our estimate would likely be a conservative measure.

6. Concluding Remarks

This study provides a comprehensive analysis of the economic impact of mass shootings on the U.S. retail sector. We apply a difference-in-differences framework

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on individual store-level transaction data to assess the economic impact of mass shootings on retail activities. We provide empirical evidence that mass shootings lead to significant and measurable reductions in retail performance and foot traffic. Mass shootings negatively affect not only the directly impacted focal stores but also stores within a 1.25-mile radius, with the effect diminishing as the distance from the incident increases. The economic cost of mass shootings is estimated to be $27 billion annually in forgone revenue to retail businesses or 0.5% of the U.S. annual retail GDP.

We conclude by discussing several implications. First, although the issue of gun violence and mass shootings has been widely studied, we bring a novel perspective by examining how consumers and individual businesses are affected by mass shootings in retail settings. Our findings suggest that the impacts on retail businesses should be included as part of the policy discussion surrounding gun violence and mass shootings.

Second, we provide insights into the mechanism driving the decline in consumer spending following mass shootings. By analyzing foot traffic data, we find decreases in both customer visits and dwell time at affected retail locations, implying the role of consumer fear and safety concerns. We also find that establishments selling nonessential goods and those requiring longer customer dwell time experience more pronounced adverse effects. Furthermore, possibly as a result of desensitization, businesses in areas perceived as less prone to gun violence experience greater disruptions from mass shooting incidents.

Lastly, online channels can serve as a tool for businesses in the aftermath of social disruptions, such as mass shootings. We find a substitution from offline to online channels after incidents among restaurants, which we attribute to consumers avoiding the affected physical locations. From a managerial perspective, having an online channel may allow businesses to recover some of their losses, making them more resilient to negative shocks.

Endnotes

[1] Gun Violence Archive, https://www.gunviolencearchive.org/.

[2] Notable incidents include the 2023 Allen Outlet Mall shooting, 25 miles north of Dallas, Texas, killing nine and injuring seven, and the 2019 Walmart mass shooting in El Paso, Texas, where 23 shoppers were killed.

[3] See www.retaildive.com/news/mall-target-retail-mass-shootings/649744/.

[4] See www.wsj.com/articles/a-new-risk-factor-in-corporate-disclosures-active-shooters-11565170202.

[5] See https://www.gunviolencearchive.org/.

[6] See https://docs.safegraph.com/docs/spend-summary-statistics.

[7] Other papers that have used SafeGraph Spend are Zheng et al. (2024), Parolin et al. (2024), and Jin et al. (2024), but to our knowledge, none had done so in marketing. Other studies in marketing use slightly different data sets. For example, Misra et al. (2022) uses a zip-code-level debit card database. Ghosh Dastidar et al. (2023) and Huang et al. (2023) use foot traffic data. Kim and McCarthy (2024) use a panel of individual-level debit and credit card transaction data.

[8] All metrics, except revenue, are available only monthly. Revenue data is collected daily and weekly and then aggregated. Table D.2 in the Online Appendix shows the robustness of the main results at daily and weekly levels.

[9] Measurements of 0.25 miles or 400 meters is the distance often used as an acceptable walking distance in urban planning to assess walkability. It is the distance an average person can walk in five minutes.

[10] The average distance an American traveled for shopping in 2022 was 5.8 miles (Abbiasov et al. 2022). Our result is also robust to using a minimum distance of 20 miles instead of 10 miles (Online Appendix D.2).

[11] SafeGraph discontinued Patterns (foot traffic) data as of January 2023. See https://docs.safegraph.com/docs/monthly-patterns for more information.

[12] Other plausible explanations include the existence of supply-side promotion (see Section 5.4), pricing factors that reduce basket size, and changes in the pool of consumers at these stores after shooting.

[13] SafeGraph tracks more than 7,700 distinct brands, including all major brands and national chains in the United States, such as Chick-fil-A, 7-Eleven, and Subway.

[14] Table 3 later demonstrates that nonessential stores experience greater revenue losses.

[15] To be more conservative, if a closed store is found to be replaced by a newly opened store, we exclude the losses associated with these closed stores and drop them from our calculation. This occurs rarely, with less than 1% of closed stores being observed to be replaced by new stores.

[16] The 2024 National Retail Federation's *The Economic Contribution of the U.S. Retail Industry.*

[17] Sit-down food places include NAICS codes 3121 (Beverage Manufacturing, such as brewery pubs), 722410 (Drinking Places (Alcoholic Beverages)), 722511 (Full-Service Restaurants), and 722515 (Snack and Nonalcoholic Beverage Bars, such as coffee shops). Fast-food restaurants include NAICS codes 722513 (Limited-Service Restaurants) and 311811 (Retail Bakeries).

[18] Non–food service retail is defined as NAICS codes starting with 44 and 45. Essential stores include the following NAICS codes: 4413 (Automotive Parts, Accessories, and Tire Stores), 4441 (Building Material and Supplies Dealers), 4451 (Grocery Stores), 4452 (Specialty Food Stores), 4453 (Beer, Wine, and Liquor Stores), 4461 (Health and Personal Care Stores), 4471 (Gasoline Stations), 4523 (General Merchandise Stores, including Warehouse Clubs and Supercenters), and 4532 (Office Supplies, Stationery, and Gift Stores). Nonessential stores comprise all other retail stores under NAICS codes 44 and 45.

[19] Note that restaurants here exclude bars and other primarily alcoholic beverages drinking places.

[20] Our geo-coded transaction data contain in-person purchases and online orders fulfilled by individual stores. However, it does not capture online orders fulfilled by central warehouses located elsewhere. Because restaurants typically fulfill online orders such as take-outs and delivery orders at individual storefront locations, our data provide a detailed view of online ordering patterns, making them suitable for studying channel substitution effects.

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